

Presynaptically Localized Cyclic GMP-Dependent Protein Kinase 1 Is a Key Determinant of Spinal Synaptic Potentiation and Pain Hypersensitivity

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Synaptic long-term potentiation (LTP) at spinal neurons directly communicating pain-specific inputs from the periphery to the brain has been proposed to serve as a trigger for pain hypersensitivity in pathological states. Previous studies have functionally implicated the NMDA receptor-NO pathway and the downstream second messenger, cGMP, in these processes. Because cGMP can broadly influence diverse ion-channels, kinases, and phosphodiesterases, pre- as well as post-synaptically, the precise identity of cGMP targets mediating spinal LTP, their mechanisms of action, and their locus in the spinal circuitry are still unclear. Here, we found that Protein Kinase G1 (PKG-I) localized presynaptically in nociceptor terminals plays an essential role in the expression of spinal LTP. Using the Cre-lox P system, we generated nociceptor-specific knockout mice lacking PKG-I specifically in presynaptic terminals of nociceptors in the spinal cord, but not in post-synaptic neurons or elsewhere (SNS-PKG-I(-/-) mice). Patch clamp recordings showed that activity-induced LTP at identified synapses between nociceptors and spinal neurons projecting to the periaqueductal grey (PAG) was completely abolished in SNS-PKG-I(-/-) mice, although basal synaptic transmission was not affected. Analyses of synaptic failure rates and paired-pulse ratios indicated a role for presynaptic PKG-I in regulating the probability of neurotransmitter release. Inositol 1,4,5-triphosphate receptor 1 and myosin light chain kinase were recruited as key phosphorylation targets of presynaptic PKG-I in nociceptive neurons. Finally, behavioural analyses in vivo showed marked defects in SNS-PKG-I(-/-) mice in several models of activity-induced nociceptive hypersensitivity, and pharmacological studies identified a clear contribution of PKG-I expressed in spinal terminals of nociceptors. Our results thus indicate that presynaptic mechanisms involving an increase in release probability from nociceptors are operational in the expression of synaptic LTP on spinal-PAG projection neurons and that PKG-I localized in presynaptic nociceptor terminals plays an essential role in this process to regulate pain sensitivity.